BARANINATHAN S P

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SUMMARY

Computer Science Engineering student with hands-on experience in Artificial Intelligence, Machine Learning, Deep Learning, and Computer Vision. Experienced in developing real-time AI solutions using Python, OpenCV, TensorFlow, YOLOv8, and Flask. Built projects involving real-time flood detection, medical image classification, and weather forecasting. Strong understanding of machine learning workflows, data preprocessing, model development, evaluation, and deployment.

EDUCATION

FRANCIS XAVIER ENGINEERING COLLEGE — 2023 – 2027
B.E - COMPUTER SCIENCE ENGINEERING — CGPA : 7.8

LITTLE FLOWER MATRICULATION HIGHER SECONDARY SCHOOL — 2022 – 2023
MATHS COMPUTER SCIENCE GROUP — HSC - Class XII

SKILLS

Languages & Web Technologies: C, Python, SQL (MySQL), HTML5, CSS3 , JavaScript, React.js, REST APIs, Flask.
Computer Vision: OpenCV, YOLOv8, Image Processing, Object Detection, Image Classification, Video Analytics
Embedded Systems & IoT: Raspberry Pi, ESP32, ESP8266, Sensor Integration, Microcontrollers, Real-Time Monitoring
Tools: Visual Studio Code, MySQL Workbench, GitHub, Grafana, Microsoft Excel, Google Colab.
Database: MySQL, XAMPP, SQL

Portfolio | Github | LinkedIn | Leetcode

PROJECTS

AI-URBAN FLOOD DETECTION SYSTEM – AI / WEB — JAN - FEB 2026

Patented

- Team Leader & Developer: Built a patented YOLOv8 computer vision system for real-time flood monitoring.
- Developed a real-time CCTV flood detection system using Python, YOLOv8, OpenCV, and Flask, deployed on Raspberry Pi 4 for 24/7 monitoring.

DISEASE DETECTION SYSTEM - AIML / WEB — FEB - APR 2026

- Team Leader & Developer: Developed a machine learning model utilizing Python, OpenCV, and TensorFlow.
- Developed a machine learning/deep learning model for automated medical image classification using Python, OpenCV, and TensorFlow.

WEATHER INTELLIGENCE FORECASTING SYSTEM - IOT — JAN - MAR 2025

- Team Leader & Developer: Engineered an award-winning IoT weather station using hardware sensors and Python.
- Built an IoT-based weather monitoring system using ESP8266 and environmental sensors.

EXPERIENCE

EDGEMATRIX MULTINATIONAL CORPORATION, MADURAI — JAN - FEB 2026

- AI/ML Intern: Developed and optimized Artificial Intelligence and machine learning solutions using Python.
- Built interactive dashboards to monitor real-time data and extract actionable insights.
- Built IoT-based monitoring solutions using ESP32, ESP8266, sensors, and Raspberry Pi.

SOFTWARE DEVELOPER, DESIGNER (FREELANCE) — PRESENT

- Full-Stack Developer & Designer: Built responsive websites utilizing HTML5, CSS3, JavaScript, and React JS.
- Created custom logos, branding mockups, and visual identities based on specific client needs.
- Managed end-to-end client coordination to gather requirements and deliver high-quality digital solutions.

AWARDS

- 1st Prize (15,000 cash) VISAI 2026 (16th International Project Competition)
- Published a Patent for AI-Urban Flood Detection System
- 1st Place – AATRAL Expo for Best Product
- 3rd Prize – Astro Aerospace Pvt. Ltd. Competition

CERTIFICATIONS

- Red Hat Linux (RH124)
- SQL by Hacker Rank
- Certified in OpenCV Bootcamp
- Internet of Things in NPTEL